

CHAPTER 1

Engineering Sociality in a Culture of Connectivity

1.1. INTRODUCTION

Meet the Alvin family. Pete is a 45-year-old biology teacher whose hobby is paragliding. He has a Facebook page, although lately he has been negligent in maintaining his network of “friends.” Through LinkedIn, Pete keeps up his professional profile and occasionally hooks up with other members from the national teachers union. An early adopter of social media, he became an enthusiastic contributor to Wikipedia in 2004, and still adds infrequent entries about his specialty, lizards, to the online encyclopedia. Pete also used to be a member of a paragliding group on YouTube, which, back in 2006, actively communicated via short videos of spectacular glides; the group later dissipated, and he only sporadically checks the site for interesting glides. Pete’s wife Sandra is a former journalist who now makes money as a freelance publicist specializing in food. She has over 8,000 followers on Twitter and keeps an elaborate blog that also serves as her personal public relations site. An active family of “netizens,” the Alvins order books via Amazon and download music via iTunes; Sandra uses Skype to have video chats with her brother in Hong Kong; their 16-year-old daughter Zara is a fanatic Facebook user—456 friends right now—and she also uses Pinterest for “pinning” and sharing photos; and their 12-year-old son Nick is a devoted gamer, who has recently discovered CityVille, a social network game developed by Zynga.

The Alvins represent a middle-class family in an average American town in the year 2012. Over the past decade, their professional and personal lives have gradually become inundated with social media platforms. Platforms

like Facebook, YouTube, Wikipedia, and many others enable people like the Alvins to make connections by sharing expressive and communicative content, building professional careers, and enjoying online social lives. In fact, the widespread presence of platforms drives people to move many of their social, cultural, and professional activities to these online environments. Teenagers like Zara Alvin cannot imagine a life without Facebook, and Sandra has become primarily dependent on Twitter for maintaining customer relations. Pete, however, has become less active on—and more critical of—the sites he used to frequent several years ago.

Now multiply the Alvins. Every single day, millions of individuals interact through social media. In December 2011, 1.2 billion users worldwide—82 percent of the world's Internet population over age 15—logged on to a social media site, up from 6 percent in 2007.¹ Within less than a decade, a new infrastructure for online sociality and creativity has emerged, penetrating every fiber of culture today. Social media, roughly defined as “a group of Internet-based applications that build on the ideological and technological foundations of Web 2.0, and that allow the creation and exchange of user-generated content” (Kaplan and Haenlein 2010: 60), form a new online layer through which people organize their lives. Today, this layer of platforms influences human interaction on an individual and community level, as well as on a larger societal level, while the worlds of online and offline are increasingly interpenetrating. Originally, the need for *connectedness* is what drove many users to these sites. When Web 2.0 first marshaled the development of so-called social media, in the early years of the new millennium, participatory culture was the buzzword that connoted the Web's potential to nurture connections, build communities, and advance democracy. Many platforms embraced this rekindled spirit when they started to make the Web “more social.”

With the rapid growth of social media platforms came the incorporation of sites by existing and new information companies. Companies often appeared less interested in communities of users than in their data—a by-product of making connections and staying connected online. *Connectivity* quickly evolved into a valuable resource as engineers found ways to code information into algorithms that helped brand a particular form of online sociality and make it profitable in online markets—serving a global market of social networking and user-generated content. Large and influential platforms such as Facebook, Twitter, YouTube, and LinkedIn exploded in terms of users and monetizing potential, alongside countless smaller profit and nonprofit sites. As a result of the interconnection of platforms, a new infrastructure emerged: an ecosystem of connective media with a few large and many small players. The transformation from networked communication to

“platformed” sociality, and from a participatory culture to a culture of connectivity, took place in a relatively short time span of ten years.

This chapter’s argument focuses not on a descriptive account of how social media affected one family, but on the need for a *critical history* of the rise of social media. Such a history is needed to comprehend current tensions in the ecosystem in which platforms and ever-larger groups of users operate. By exploring technical, social, economic, and cultural perspectives on social media, we can elucidate how recent changes in our global media landscape have profoundly affected—if not driven—our experience of sociality.

1.2. FROM NETWORKED COMMUNICATION TO PLATFORMED SOCIALITY

The invention of the World Wide Web in 1991, when Tim Berners-Lee managed to connect hypertext technology to the Internet, formed the basis of a new type of networked communication. Weblogs, list-servers, and e-mail services helped form online communities or support offline groups. Until the turn of the millennium, networked media were mostly generic services that you could join or actively utilize to build groups, but the service itself would not automatically connect you to others. With the advent of Web 2.0, shortly after the turn of the millennium, online services shifted from offering channels for networked communication to becoming interactive, two-way vehicles for networked sociality (Castells 2007; Manovich 2009). These new services, which opened up a myriad of possibilities for online connections, were initially perceived as a new global infrastructure, like water pipes or electricity cables, analogues to the Web itself.

It is a truism to say that media have historically coevolved with the public that uses them, as well as with the larger economy of inscription. The world’s complex constellations of media, in the view of Lisa Gitelman, should be conceived as the “socially realized structures of communication, where structures include both technological forms and their associated protocols, and where communication is a cultural practice, a ritualized collocation of different people on the same mental map, sharing or engaged with popular ontologies of representation” (2008: 7). Over the past two centuries, media technologies matured as part of everyday social practices. Generic technologies like the telephone and the telegraph developed in conjunction with communicative routines or cultural practices, such as chatting on the phone or sending short messages over the wire. As a medium coevolves with its quotidian users’ tactics, it contributes to shaping people’s

everyday life, while at the same time this mediated sociality becomes part of society's institutional fabric. Media histories and archaeologies provide ample evidence of this complex coevolution, relating technologies to users and organizations to infrastructures (Winston 1998; Kittler 1999; Zielinski 1999; Marvin 1988).

With Web 2.0 maturing into a functional infrastructure, users moved more of their everyday activities to online environments; these activities were not simply channeled by platforms, but *programmed* with a specific objective. This move shifted the emphasis from providing a utility to providing a customized service—a transformation akin to the change from delivering water through pipelines to distributing bottled Evian water or to a water-filtering system. Whereas before, websites were generally operated as *conduits* for social activity, the new platforms increasingly turn these conduits into *applied services*, rendering the Internet easier to use but more difficult to tinker with. Social media platforms, as they are now commonly called, epitomize the larger conversion from all-purpose devices to linear applied services—a development that Jonathan Zittrain (2008: 104–7) has persuasively touted as “appliancization.” When companies started to build their platforms on the generic Web 2.0 infrastructure, they often presented themselves as utilities transmitting communication and information data. But even if many big platforms still want people to think of them as such, this layer of applied platforms is anything but a neutral utility exploiting a generic resource (data): they built on the “ideological and technological” foundations of Web 2.0, as Kaplan and Haenlein suggest in the definition quoted above.

Indeed, most Web 2.0 platforms started out as indeterminate services for the exchange of communicative or creative content among friends. These services often emanated from community-bound initiatives—a group of college students, photo aficionados, video enthusiasts—who adopted a specific niche of online interaction and developed a mediated routine practice. It is a common fallacy, though, to think of platforms as merely *facilitating* networking activities; instead, the construction of platforms and social practices is mutually constitutive. Sociality and creativity happen while people are busy living their lives. Michel de Certeau, in *The Practice of Everyday Life* (1984), proposes that people use *tactics to negotiate* the strategies that are arranged for them by organizations or institutions. That is precisely what happened with the development of social media platforms and the apps built on top of them: users “negotiate” whether and how to appropriate them in their quotidian habits.

Many of the habits that have recently become permeated by social media platforms used to be informal and ephemeral manifestations of social life.

Talking to friends, exchanging gossip, showing holiday pictures, scribbling notes, checking on a friend's well-being, or watching a neighbor's home video used to be casual, evanescent (speech) acts, commonly shared only with selected individuals. A major change is that through social media, these casual speech acts have turned into formalized inscriptions, which, once embedded in the larger economy of wider publics, take on a different value. Utterances previously expressed offhandedly are now released into a public domain where they can have far-reaching and long-lasting effects. Social media platforms have unquestionably altered the nature of private and public communication.

From the late 1990s onward, Blogger (1999), Wikipedia (2001), Myspace (2003), Facebook (2004), Flickr (2004), YouTube (2005), Twitter (2006), and a wide array of ensuing platforms began to offer web tools that sparked old and new online communication tactics. Most organizations operating these platforms aimed at penetrating a particular online activity with their coding technologies, and, ideally, their brand name would become the marker for a specific mediated activity. Brands such as Twitter, YouTube, MSN, and Skype have become synonyms for microblogging, video sharing, chatting, and videoconferencing—novel communicative interactions these platforms either co-developed or helped redesign. The pinnacle of a company's success in permeating a social activity is when a brand turns into a verb. The earliest example of such coding and branding phenomena in the online world is the evolution of “googling,” now a synonym for online search. Googling, following Gitelman's definition above, could be called a “ritualized collocation” that developed in a “larger economy of inscription.” Online searching—for example, looking up the meaning of a word, checking for the latest movies, or finding a specific scholarly source—has become part of an everyday routine. Simultaneously, this routine nested itself in the heart of a larger online economy of inscription, where search engines form the valves of content distribution. Few platforms have reached the stage where their brand has turned into a verb; at this point in time, “skyping” and “tweeting” perhaps come closest.²

Evidently, social media platforms, rather than being finished products, are dynamic objects that are tweaked in response to their users' needs and their owners' objectives, but also in reaction to competing platforms and the larger technological and economic infrastructure through which they develop (Feenberg 2009). In the year 2000, the Web that would come to sustain online sociality and creativity was still a vast unexplored territory, where boundaries between different mediated activities had yet to be demarcated. It was a new frontier, a bonanza where rules and laws from the “old” territories no longer applied and new ones had not crystallized yet.

The earliest cultivators of this new land were search engines, browsers, and web directories; of the many search engines that sprang up around the turn of the millennium, Google Search—including its many specialized services—has emerged victorious, leaving a few small engines trailing behind.³ Like web browsers, search engines tend not to be presented as applications built to search, navigate, and connect information on the WWW, but they are conspicuously equated to the Web itself.⁴ Over the past decade, there has been an unprecedented proliferation of social media platforms as each one of them tried to occupy the largest possible chunk of this new terrain. Whereas some have succeeded (Facebook, YouTube), others have waxed and waned (Flickr, Myspace), and yet others have quietly disappeared (remember Xanga?). On top of this layer, millions of application program interfaces (APIs) and services have been built that depend on the services of Facebook, Google, Twitter, and so on, for their success, and new ones emerge every day. The entire ecosystem of interconnected platforms and applications has been in flux and will remain volatile for some time to come.

While it would be virtually impossible to inventory all platforms and their individual evolutions, it makes analytical sense to distinguish various *types* of social media. A major type involves what is called “social network sites” (SNSs). These sites primarily promote interpersonal contact, whether between individuals or groups; they forge personal, professional, or geographical connections and encourage weak ties. Examples are Facebook, Twitter, LinkedIn, Google+, and Foursquare. A second category concerns sites for “user-generated content” (UGC): they support creativity, foreground cultural activity, and promote the exchange of amateur or professional content. Well-known UGC sites are YouTube, Flickr, Myspace, GarageBand, and Wikipedia. On top of these, we can add the category of trading and marketing sites (TMSs): these sites principally aim at exchanging products or selling them. Amazon, eBay, Groupon, and Craigslist come to mind as noteworthy examples. Another distinctive category consists of play and game sites (PGS), a flourishing genre with popular games such as FarmVille, CityVille, The Sims Social, Word Feud, and Angry Birds. This classification of social media platforms is far from exhaustive, and integrating the various types into a single book-length argument would be undoable. For this reason, I will focus primarily on SNS and UGC sites here as the main grounds on which online sociality and creativity have developed.

Important to add here is that there are no sharp boundaries between various platform categories because carving out and appropriating one or more specific niches is part of the continuous battle to dominate a segment of online sociality. Facebook, whose prime target is to promote social networking, also encourages its users to add creative products such as photos

or short videos. YouTube, a site primed to generate creative content by users, can also be considered a social network site because communities share specific postings (e.g., anime videos). Despite Google's keen attempts to turn YouTube into an SNS, it has remained primarily a site for UGC, prompting the search company to start its own social networking service, Google+, in May 2011. Meanwhile, Facebook and Google try to expand their existing platforms with commercial and game services through partnerships and takeovers, making them also major players in the TMS and PGS branches.

Sharply delineating various types of social media platforms is impossible, and yet identifying their objectives is key to understanding how platforms build different niches of sociality and creativity or, for that matter, commerce or entertainment. What we have seen over the past ten years is that many platforms started out in one particular domain (e.g., online search or social networking) and gradually encroached upon each other's territory while trying to contain users inside their own fenced-off turf. Therefore, it is instructive to track how a few rapidly growing platforms began to dominate online sociality, occupying as many niches as possible. Google and Facebook each conquered a sizable chunk of this layer, to such an extent that new developers are increasingly dependent on these platforms for building new applications. We can only gain insight into the mutual shaping of platforms and apps if we view them as part of a larger online structure where every single tweak affects another part of the system. Or, to put it more in general terms, the online ecosystem is embedded in a larger sociocultural and political-economic context where it is inevitably molded by historical circumstances.

1.3. MAKING THE WEB SOCIAL: CODING HUMAN CONNECTIONS

To get a better sense of the ecosystem's emergence, we need to go back a bit further in history. In the early 1970s, computers and information technology had a dubitable reputation as instruments of control, mostly wielded by Orwellian bureaucratic governments or by giant corporations. The counterculture, born in the 1960s and matured in the early 1970s, paired values of community and collectivity with the imperative of personal freedom and empowerment—values that clashed with lingering associations of oppression and compromised individuality still hovering around information technology. It was not until the late 1970s when computers began to be seen as potential instruments of liberation rather than oppression. In a lucid account of the gradual convergence of the counterculture with “geek”